Tutorial and Files Explained

Car Project/DATA/CAR.TIM - This is an image file made using the TIMTOOL found in "C:/psyq/bin/TIMTOOL".

/Car Project/DATA/GRID.TMD and /Car Project/DATA/CAR.TMD - These are model files. To convert a model to TMD, first you need to install Blender, then install a plugin to export the model in the right format.

How to install Blender and the export plugin

1. open "blender-3.2.1-windows-x64" and install

2. Open Blender and go to Edit --> Preferences and click the "Add-ons" tab. Then in the top right of the window click "Install".

3. Browse to the file "Playstation RSD Exporter.py" and click "Install Add-on"

4. Now type RSD into the search bar (it's right under the install button), then make sure the checkbox is checked next to the newly installed plugin.

5. You can now close the Preferences window. The included “Materials Example Project.blend” file can be placed in Blender’s asset library folder and can be accessed via Blender’s “Asset Browser” panel. This file contains various material options you can use as a starting point that are supported by the RSD exporter.

Download the latest version of this plugin and asset library here: https://github.com/rubixcube6/PlayStation-RSD-exporter-for-Blender-3.2.1

Now we need to convert a model to RSD format. (This will be converted to TMD later) RSD is a human readable format meaning that if there is any issues with a texture name or something else it is easy to debug and edit in a text editor. TMD is a binary format which is not editable in a text editor but is readable by the PlayStation.

How to convert a model to RSD in Blender

1. Open the /Car Project/DATA/Source Files/CAR.blend file in Blender.

2. Go to File --> Export --> RSD (PlayStation) Export

3. Browse to where you want to export it, Name it what you want, and click "Export RSD". It will export 3 files.

Now we need to convert the texture image from your model to TIM using the TIMTOOL found in "C:\psyq\bin\TIMTOOL".

How to use the TIM tool

1. File ---> Import Image Files

2. Find and select your image and make sure "Import Colour Depth" is set to 8bit and make sure the checkbox for "Set for all colours except Black" and "Set for Black" are un-checked then click OK. This makes sure that all pixels set to rgb 0, 0, 0 (absolute black) are transparent. (It will place a TIM on your desktop but you're not done yet)

3. Drag your image on the left to somewhere in the blue-ish/green-ish area. Just make sure it does not overlap the 2 buffer areas. I placed mine at X: 640 Y: 0. (That is set under "TIM Information Pixel Information)

4. Under CLUT Information set this somewhere else in the blue/green area so it doesn't overlap your main image. This is just your image's color pallet. I set mine to X: 0 Y: 511. (That's the small area under the buffer areas)

5. Tim Options ---> Save Modified TIM's (It will save in the same folder as your original image)

6. Drag your TIM files to the data folder in your project. (Be sure to update your mkpsxiso/cuesheet.xml to include the new TIM files)

7. Also, copy the file and paste it next to your newly exported RSD model.

That's how to prepare a TIM file. You can add more images to the green/blue area for multiple textures. Just make sure Nothing is overlaps.

Now for the grand finally!

Converting your RSD model to TMD

1. Drag the following files into the "RSD 2 TMD" folder.

.RSD
.PLY
.MAT
.TIM - All tim textures associated with the model

2. Open the "Convert.bat"

3. It will ask for a file name. Type it in and hit Enter. No need to include the file extension.

4. Drag your new TMD file and it's TIM texture to the data folder in your project. (Be sure to update your mkpsxiso/cuesheet.xml to include the new TMD model and TIM Texture file)

File Information

build iso and run.BAT - This will automatically compile your game and uses what is in the mkpsxiso folder to make an ISO. Then it will open the iso in an emulator. Edit this bat file and replace the path to the emulator with your own.

mkpsxiso/cuesheet.xml - This tells the mkpsxiso program what files to put on the CD. Add your TIM files here.

SYSTEM, MAKEFILE.mak, and LICENSEA.DAT - These are just necessary files for making an iso. No need to edit anything here for this project.

MAIN.C - This is the file that gets compiled first by build iso and run.BAT

You don't really need to understand all of the following to get something working.

Constants.h - A bunch of necessary code that does some of the heavy lifting.

Controller.c - Code that makes it easier to program input from the controller.

2D.c - Functions that relate to making a 2D game.

3D.c - Functions that relate to making a 3D game

Audio.c - Functions that handle playing Audio

ReadCD.c - Functions that handle CD loading

IMPORTAINT!

If it won't compile, you may need to edit **C:\psyq\bin\PSYQ.INI** and add **libds.lib** to the line that says stdlib=

If that still does not work just replace that whole line with the following.

stdlib=libgs.lib libgte.lib libgpu.lib libspu.lib libsnd.lib libetc.lib libapi.lib libsn.lib libc.lib libcd.lib libcard.lib libmath.lib libds.lib libapi.lib